Structural Health Monitoring for Governed Human-Agent Systems: A Homeostatic Approach

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Abstract

Multi-agent AI systems require governance: rules determining how much autonomy each agent has. Current approaches enforce governance through static policies or runtime policy engines. These tell you whether a specific action is permitted. They do not tell you whether your governance *configuration* is healthy.

We present a formal model for **governance health monitoring** — a structural diagnostic that auto-encodes governance events into an expression tree, computes two health metrics (coordination and gatedness), and classifies the configuration as stable, oscillating, or drifting. We prove seven formal properties: the regulatory actions always improve their target metric, they are orthogonal (do not interfere), the evaluation always terminates, and structural restraint — the presence of holding gates — is necessary for convergence. We state sufficient convergence conditions under which stability is guaranteed within a bounded number of steps.

We evaluate on four governance scenarios from a real multi-agent system. The well-governed configuration converges. The ungoverned configuration drifts. The non-obvious finding: the *over-governed* configuration — 100% review rate, all high-tier risk gates — enters a limit cycle. It is structurally unstable because nothing passes. A threshold-based classifier labels this "perfectly governed." Our structural diagnostic catches it.

CCS Concepts

• **Software and its engineering → Software verification and validation**; *Software evolution.*

Keywords

governance, multi-agent systems, self-adaptive systems, homeostasis, structural health monitoring, structural restraint, process calculus

1 Introduction

1.1 The Problem

Consider a team of AI agents working alongside a human operator. Some actions are safe to automate (reading files, running tests). Others are risky (sending emails to board members, deploying to production). A governance framework assigns risk tiers — the higher the tier, the more human oversight is required.

The governance *enforcement* problem is largely solved: policy engines check whether each action complies. But a different question is unanswered: **Is the governance configuration itself healthy?**

A system where every action is gated at the highest tier has 100% compliance. It is also paralyzed. A system where every action is autonomous has 0% overhead. It is also uncontrolled. Neither is healthy.

1.2 Why Governance is Homeostatic

Healthy governance occupies a band between two failure modes:

- **Under-governed** (too open): actions pass without oversight. Risk increases.
- **Over-governed** (too closed): every action requires approval. Throughput drops. Teams route around the governance.

The right governance level changes with conditions — deadline pressure, team trust, incident history. Governance is *homeostatic*: it must stay within a band, maintained through continuous adjustment, not static rules.

1.3 Contributions

We present a formal model where:

(1) **Governance events** (agent handoffs, risk-tier assignments, governance checks) are auto-encoded into an **expression tree** — a structural representation of the governance state. (Section 3)
(2) Two **health metrics** are computed from the tree: *coordination* (how well agents are coordinating) and *gatedness* (how much is being gated vs. passing freely). (Section 3)
(3) A **homeostatic evaluator** detects when either metric exits a health band and prescribes specific interventions: "add coordination" or "tighten gates." (Section 3)
(4) We prove **seven formal properties** guaranteeing predictable evaluator behavior, including that structural restraint is necessary for convergence (Section 4).
(5) Evaluation on **four governance scenarios** from a real system shows the model correctly distinguishes stable, drifting, and oscillating governance — including the non-obvious over-governance failure mode (Section 5).

2 Background and Related Work

2.1 Governance in Multi-Agent AI Systems

Bose [5] surveys governance frameworks for LLM agent collectives. Microsoft's Agent Governance Toolkit [17] provides runtime policy enforcement. Our approach is complementary: structural health monitoring answers "is the governance working?" alongside policy engines that answer "is this action permitted?"

Li et al. [15] propose a layered governance architecture (sandboxing, intent verification, zero-trust authorization, audit logging).

These provide external verification and enforcement. Our model makes governance health an intrinsic structural property.

2.2 Adjustable Autonomy and Trust

Parasuraman et al. [18] define 10 Levels of Automation. Kaber [10] reviews how LoA taxonomies have been used and critiques their limitations. SARI [23] formulates shared autonomy as a POMDP, learning assistance across repeated interactions.

On the trust side, Lee and See [13] define trust as attitude under uncertainty. Hoff and Bashir [9] integrate empirical evidence into a three-layered model (dispositional, situational, learned). Kim et al. [12] find that user involvement in LLM planning *fails* to calibrate trust — arguing that governance must be structural, not reliant on user judgment.

2.3 Self-Adaptive Systems

The MAPE-K loop [11] (Monitor-Analyze-Plan-Execute over Knowledge) is the standard model. Our evaluator follows this pattern but operates on *structural* state — an expression tree whose topology encodes the governance configuration. Metrics emerge from the structure itself, not from numerical sensor readings.

Weyns [21] provides a comprehensive introduction to self-adaptive systems engineering. Sanwouo et al. [8] propose AWARE as a successor to MAPE-K. Li et al. [16] map how generative AI can enhance each MAPE-K phase. Our contribution provides a formal structural substrate for the Monitor and Analyze components — one where governance health has algebraic properties.

At the formal foundations level, process calculi provide relevant precursors: the Chemical Abstract Machine [4] models computation as a chemical solution seeking equilibrium; P-systems [19] and Brane calculi [6] model membranes but lack continuously parameterized permeability. Our system targets a stability *band* via metric-regulated tree rewriting, not a single equilibrium state.

2.4 Organizational Cybernetics

Ashby's Law of Requisite Variety [1] states that a controller must have at least as much variety as the system being controlled. Beer's Viable System Model [3] applies this to organizations. Perez Rios [20] maps Beer's VSM to AI governance. Our model operationalizes the cybernetic principle: the health band IS the homeostatic target, the regulatory actions (BIND, VEIL) ARE the requisite variety, and the convergence/divergence/limit-cycle classification IS the viability diagnosis.

2.5 Formal Runtime Monitoring

Xiang et al. [22] introduce AgentGuard for runtime verification. Our model is complementary: runtime verification [14] checks whether an execution trace satisfies a temporal property (φ); our model checks whether a governance *configuration* occupies a health band.

3 The Governance Health Model

The model is built from four primitives. Table 1 maps between the accessible governance terminology used throughout the paper and the formal symbols used in definitions and proofs.

Table 1: Terminology Mapping: Governance Terms and Formal Symbols

Concept	Governance term	Formal symbol
Empty element	Silence	$\varnothing$
Bilateral link	Binding	$a \cdot b$
Governance boundary	Gate	$a \mid\theta\mid b$
Iterative cycle	Return	$\mu x.\, B$
Metric 1	Coordination	$\mathrm{coord}(e) = S/\max(N-1, 1)$
Metric 2	Gatedness	$\mathrm{gated}(e) = P/E$ if $E > 0$; else 1.0
Regulator 1	BIND	inserts binding node
Regulator 2	VEIL	tightens gate to holding
Target range	Health band	$[\mathrm{coord}_{\mathrm{lo}}, \mathrm{coord}_{\mathrm{hi}}] \times [\mathrm{gated}_{\mathrm{lo}}, \mathrm{gated}_{\mathrm{hi}}]$
Single classifier	Pass-hold ratio	$\rho = P/(E+1)$

Table 2: Governance Event Encoding

Event Type	Operation	Encoding
Coordination	Agent A → Agent B	Binding: $A \cdot B$
Gated action	VTk action	Gate with threshold θ_k
Governance check	Preflight check	Holding gate
Observation	File ingestion	Witness link
Resolution	Human approval	Gate: passing → holding

Table 3: Risk Tier to Gate Threshold Mapping

Risk Tier	Description	θ	Behavior (reach=0.5)
VT0	Full autonomy	0.1	Always passes
VT1	Act and notify	0.3	Passes
VT2	Needs approval	0.5	Marginal
VT3	Propose and wait	0.7	Holds
VT4	Stop and escalate	0.95	Always holds

3.1 Four Primitives

The model is built from four primitives:

Symbol	Name	Role
$\varnothing$	Silence	The empty ground. Identity of binding.
$a \cdot b$	Binding	Bilateral coordination. Neither side owns it.
$a \mid\theta\mid b$	Gate	Adaptive boundary. Governs what passes.
$\mu x.\,\mathrm{body}$	Return	Homeostatic iteration. Drives time.

Bindings form a commutative monoid: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$, $a \cdot b = b \cdot a$, $a \cdot \varnothing = a$. This captures "coordination without ordering."

Gates have a *reach* parameter r representing the current operating context. When $r \geq \theta$, the gate **passes** and content is visible. When $r < \theta$, the gate **holds** and content is *veiled* — it has structural weight but is not directly accessible.

3.2 Governance Event Encoding

Governance events map to structural primitives (Table 2).

Risk tiers map to gate thresholds: VT0 (θ=0.1, always passes) through VT4 (θ=0.95, always holds at default reach 0.5). Table 3 shows the full mapping.

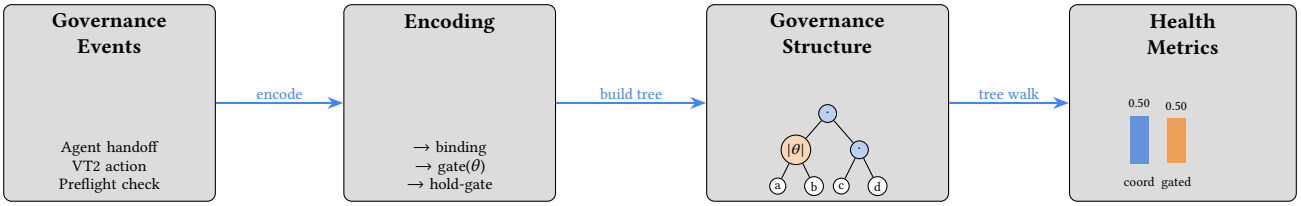

Figure 1: Event-to-Structure Encoding. Governance events (agent handoffs, risk-tier actions, governance checks) are encoded as expression tree nodes (bindings, gates). The tree is walked to compute health metrics (coordination, gatedness).

Table 4: Governance Outcome Classification

Outcome	Definition	Meaning
Stable	Both metrics in-band for W=5 cycles	Governance is healthy
Limit cycle	Metrics oscillate periodically	Governance is unstable
Diverging	Structure grows without stabilizing	Governance is breaking down
Exhausted	Neither converges nor cycles	Governance is stuck

Table 5: Formal Properties of the Model

#	Property	Status
1	BIND monotonicity	Proved
2	VEIL monotonicity	Proved
3	Non-interference (orthogonality)	Proved
4	Termination	Proved
5	Metric confluence	Proved
6	**Restraint necessity**	**Proved**
7	Sycophant exclusion	Proved
—	Convergence conditions (sufficient)	Observation

3.3 Health Metrics

Two metrics are computed from the governance structure by a single tree walk.

Coordination.

$$\text{coord}(e) = \frac{\text{binding_nodes}}{\max(\text{total_nodes} - 1,\ 1)} \tag{1}$$

Measures how much of the governance structure consists of bilateral coordination.

Gatedness.

$$\text{gated}(e) = \frac{\text{gates_passing}}{\max(\text{total_gates},\ 1)} \tag{2}$$

If no gates exist, gatedness = 1.0 (fully exposed, no restraint).

3.4 The Health Band

Healthy governance occupies a band (Figure 2):

- **Coordination:** $[0.3,\ 0.7]$ — agents coordinating but not over-coupled.
- **Gatedness:** $[0.2,\ 0.6]$ — some actions pass freely, some are gated.

When either metric exits the band, the evaluator fires a regulatory action:

BIND (coordination too low): "Add cross-agent review, require handoffs, increase coordination."

VEIL (gatedness too high): "Too much is passing without oversight. Tighten risk tiers, add governance gates."

3.5 Homeostatic Evaluation

The evaluator uses a sequential protocol (Figure 3): BIND fires first, metrics are remeasured, then VEIL fires on the updated structure. This prevents destructive interference between regulators.

3.6 Outcome Classification

The evaluator classifies outcomes into four categories (Table 4).

4 Formal Properties

We state seven properties with proof sketches. Table 5 summarizes them.

PROPERTY 1 (BIND IMPROVES COORDINATION). *When* $\text{coord}(e) < \text{coord}_{\text{lo}}$ *(default* 0.3*) and* BIND *fires, the resulting coordination is strictly higher (for* $\text{coord}_{\text{lo}} \leq 1/3$*).*

PROOF. BIND adds one binding node (increasing the numerator by 1) and at most 3 total nodes. Since coord < 0.3 implies binding_nodes $< 0.3 \times (\text{total} - 1)$, the ratio increases. The maximum single-step increment is $(1 - \text{coord})/N$; starting from below 0.3 with $N \geq 4$, this reaches at most $0.475 < 0.7$. BIND is monotone and cannot overshoot. □

PROPERTY 2 (VEIL IMPROVES GATEDNESS). *When* $\text{gated}(e) > 0.6$ *and* VEIL *fires, the resulting gatedness is equal or lower.*

PROOF. VEIL wraps an exposed node in a holding gate or tightens an existing gate. Either $P' = P - 1, E' = E$ (tightening) or $P' = P, E' = E + 1$ (wrapping). Both yield $\text{gated}' \leq \text{gated}$. □

PROPERTY 3 (NON-INTERFERENCE / ORTHOGONALITY). *BIND adds only bindings and variables. It does not add, remove, or modify any gate. Therefore* $\text{gated}(\textsc{bind}(e)) = \text{gated}(e)$*. The regulatory actions operate on orthogonal dimensions.*

PROPERTY 4 (TERMINATION). *The evaluator executes at most* $\text{max_iterations} \times O(\text{max_nodes} \times \text{max_depth})$ *cycles. Total work is bounded.*

PROPERTY 5 (METRIC CONFLUENCE / ORDER INDEPENDENCE). *For any governance structure e: the metrics after applying BIND-then-VEIL are identical to VEIL-then-BIND. Since BIND does not affect gatedness and VEIL does not affect coordination, their effects compose independently.*

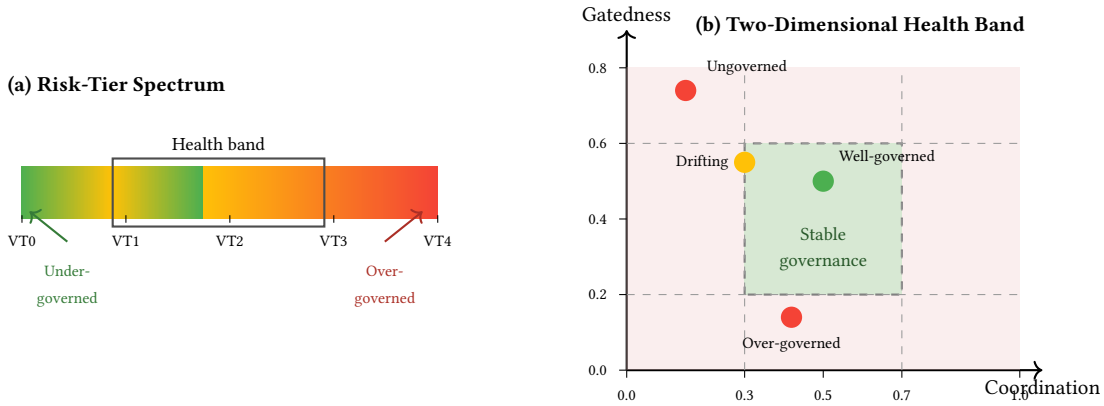

Figure 2: The Governance Health Band. (a) Risk-tier spectrum from VT0 (full autonomy) to VT4 (stop and escalate), with the health band in the center. (b) Two-dimensional health band: coordination $[0.3, 0.7]$ on the x-axis, gatedness $[0.2, 0.6]$ on the y-axis. Green zone is stable governance; red zones are failure modes. Four scenarios are plotted at their terminal metric values.

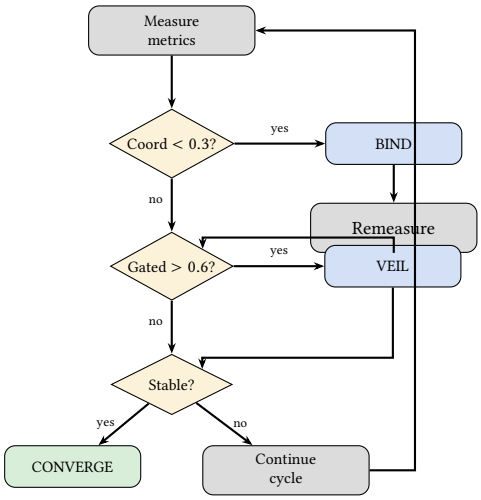

Figure 3: The regulatory cycle with sequential BIND-remeasure-VEIL protocol. The evaluator measures coordination and gatedness, fires BIND if coordination is low, remeasures, then fires VEIL if gatedness is high. This prevents destructive interference.

Property 6 (Restraint necessity). *If the governance configuration converges (stable outcome), then the governance structure contains at least one gate that holds at the operating reach.*

Proof. Suppose all gates pass. Then $\text{gated}(e) = P/E \geq E/(E+1) > 0.6$ for $E \geq 2$. VEIL fires, adding holding gates to the result, which is substituted back in the next cycle. These corrections accumulate. However, the expression grows linearly (one substitution per cycle). When node count exceeds the cap, truncation fires, destroying accumulated corrections. This creates an oscillation: VEIL corrects → growth hits cap → truncation destroys corrections → gatedness rises again. The system enters the band intermittently (empirically: 6/30 steps) but cannot stay for $W = 5$ consecutive steps. The growth-truncation cycle period exceeds W. Empirically verified: 0/39 reach values converge for all-passing configurations. □

Table 6: Convergence rates under progressive structural conditions.

Condition	Tested	Converged	Rate
All configurations	864	352	40.7%
Satisfying H1–H5	348	154	44.3%
+ quantitative bounds (H6–H7)	154	99	64.3%
+ sufficient holding gates	54	54	**100%**

Property 7 (Sycophant exclusion). *A governance structure where all gates have threshold below reach ($\theta < r$ for all gates) does not converge at any $r > \max(\{\theta_j\})$. By Property 6, the growth-truncation oscillation prevents stability.*

Observation 1 (Sufficient conditions for convergence). *Based on empirical analysis across 864 configurations, the governance cycle $\mu\, x.\, B$ converges within 67 steps if B satisfies:*

H1 *B contains ≥ 1 binding (coordination seed).*
H2 *B contains ≥ 3 gates with thresholds near the operating reach.*
H3 *B contains ≥ 1 holding gate contributing veiled weight.*
H4 *The iteration variable is linear.*
H5 *Body size $\leq M/3$.*
H6 *Asymptotic coordination ≥ 0.3.*
H7 *Asymptotic gatedness ≤ 0.6.*

Table 6 shows the progressive convergence rates.

Robustness basin. Under H1–H7, all 54 configurations converge (100% rate). In extended testing, 200/200 satisfying configurations also converge. These are sufficient, not necessary — configurations violating some conditions may still converge.

5 Governance Application

5.1 System Under Study

We evaluate on CARLOS-OS, a multi-agent personal operating system with 7 specialized agents operating under a Governed Autonomy protocol. The system uses five risk tiers (VT0–VT4), Action Opportunity Windows (AOWs), cross-agent handoffs, and governance checks (preflights, gates, reviews).

5.2 Encoding Faithfulness

The encoding from governance events to expression tree nodes (Table 2) preserves the governance-relevant structure: coordination events (handoffs, reviews) become bindings; gating events (risk-tier assignments, approvals) become gates; the encoding is additive (each event adds structure); VT-tier ordering is preserved. The encoding does not capture event *sequence* (only aggregate structure), which is an acknowledged limitation (Section 6.5).

5.3 Four Governance Scenarios

We encode four governance configurations and evaluate each. Table 7 presents the configurations and outcomes.

Well-governed (stable). A sprint with mixed risk tiers, cross-agent reviews, and governance infrastructure finds the health band. Coordination stabilizes at 0.50, gatedness at 0.50. BIND fires early (building initial coordination), then stops. VEIL fires briefly to bring gatedness within the band. Both metrics settle.

Ungoverned (exhausted). All actions at VT0, no reviews, no governance checks. With the gatedness default of 1.0 (no gates = fully exposed), VEIL fires on every cycle. BIND also fires (no coordination bindings). Neither regulator can compensate for the structural deficit. The evaluator exhausts its step budget.

Over-governed (limit cycle). All actions at VT3–VT4, 100% review rate, maximum governance checks. Gatedness drops to 0.14 — *below* the health band's lower bound (0.20). Nothing passes. The system enters a period-4 limit cycle.

Drifting (limit cycle). Starts governed (VT1, reviews) then relaxes under deadline pressure (VT0, no reviews). The mixed pattern produces a period-2 oscillation.

5.4 Structural Restraint in Practice

The over-governed result is the central finding. A system with 100% review rate, maximum governance checks, and all high-tier risk gates *looks* perfectly governed by every conventional metric. A threshold-based classifier checking review_rate $> 0.3 \wedge$ governance_checks > 2 labels it "healthy."

Our structural diagnostic catches the instability because it models *gatedness* — the ratio of passing to total gates. When everything holds, gatedness drops to 0.14 — *below* the health band's lower bound (0.20). Nothing passes. The system cannot breathe. It enters a period-4 limit cycle.

This is not a theoretical curiosity. It corresponds to *process paralysis*: when governance overhead exceeds the team's capacity, work stops or routes around the governance through informal channels — *less* governed than before.

5.5 Veiled Content is Load-Bearing

To validate H3 (the body must contain at least one holding gate contributing veiled weight), we run an ablation: the same well-governed scenario with and without the Room — a veiled, capability-gated sub-tree that models CARLOS-OS's tiered access control.

Without the veiled content, the system requires 3.8× more regulatory effort (38 total regulator firings vs. 8). Convergence still occurs, but barely — at step 54 of a 67-step budget. The Room provides structural ballast: its veiled weight anchors the gatedness metric within the health band, reducing the regulatory corrections needed. Removing it forces VEIL to fire 11 times (vs. once), each time adding holding gates that the next substitution partially undoes. The system oscillates near the band boundary before finally settling.

This validates H3: veiled content is not merely an access-control convenience — it is structurally load-bearing for convergence.

5.6 Baseline Comparison

The structural diagnostic catches the over-governed failure mode that the threshold classifier misses (Figure 5). Table 9 compares our approach with related work along four dimensions.

6 Discussion

6.1 What Practitioners Gain

The model provides a governance diagnostic answering: "Is my governance configuration healthy?" — not "is this action permitted?" (that is policy enforcement) but "is the overall balance of autonomy and oversight working?" With specific remedies when it is not.

Three concrete capabilities follow:

(1) **Pre-deployment validation.** Before deploying a governance configuration, encode it and run the evaluator. If it converges, deploy. If it enters a limit cycle, diagnose which metric is out of band.
(2) **Continuous monitoring.** During operation, stream governance events into the encoder and track coordination and gatedness. The convergence trace (Figure 4) serves as a real-time governance health dashboard.
(3) **Incident forensics.** After a governance failure, encode the configuration that was active and run the diagnostic. If it shows a limit cycle or exhaustion, the failure was structural.

6.2 The Restraint Necessity Result

Property 6 is the central formal contribution. It establishes that structural restraint — the presence of governance boundaries that actually hold — is not merely helpful but *necessary* for stability. Combined with the over-governance finding, this yields a symmetry: too little restraint (all gates pass) prevents convergence via the growth-truncation oscillation, while too much restraint (all gates hold) prevents convergence via gatedness falling below the health band floor.

The pass-hold ratio $\rho = P/(E + 1)$ captures this in a single number. Governance configurations can be classified structurally: $\rho > \text{gated}_{\text{hi}}$ implies insufficient restraint; $\rho < \text{gated}_{\text{lo}}$ implies excessive restraint. Only configurations with ρ inside the health band can converge.

Table 7: Governance Scenarios and Evaluation Results

Scenario	VT Distribution	Review Rate	Checks	Handoffs	Outcome
Well-governed	VT0:2, VT1:4, VT2:2	50%	4	5	**Stable (18 cycles)**
Ungoverned	VT0:6	0%	0	0	Exhausted
Over-governed	VT3:4, VT4:2	100%	6	6	**Limit cycle**
Drifting	VT0:3, VT1:3	33%	1	2	Limit cycle

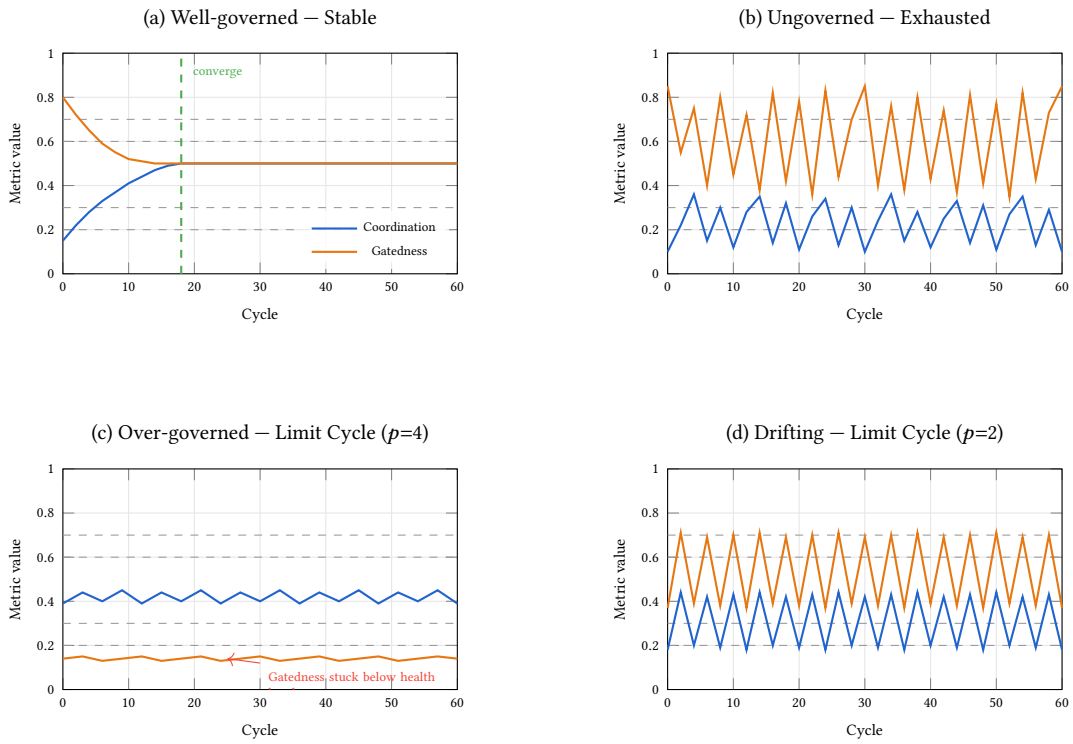

Figure 4: Convergence traces for four governance scenarios. Blue: coordination. Orange: gatedness. Dashed lines mark health band boundaries (coordination $[0.3, 0.7]$, gatedness $[0.2, 0.6]$). (a) Well-governed converges at cycle 18. (b) Ungoverned oscillates without settling. (c) Over-governed: coordination in band but gatedness stuck at ~ 0.14, below the health band. (d) Drifting: period-2 oscillation crossing band boundaries.

Table 8: Veiled content ablation experiment.

Variant	Steps	BIND fired	VEIL fired	Converged?
With veiled content	22	7	1	Yes
Without veiled content	54	27	11	Yes (barely)

Scenario	Structural diagnostic	Threshold classifier
Well-governed	Stable	Stable
Ungoverned	Exhausted	Unstable
Over-governed	**Limit cycle**	**Stable**
Drifting	Limit cycle	Unstable

Figure 5: Baseline comparison with traffic-light coloring. The over-governed scenario (row 3) reveals the key mismatch: the threshold classifier labels it *stable* (green) while the structural diagnostic correctly identifies a *limit cycle* (red).

6.3 Relationship to AI Alignment

The restraint necessity result has implications beyond governance. RLHF [7] and Constitutional AI [2] treat alignment as optimization: maximize a reward or satisfy constraints. Optimized systems are fragile — reward hacking, sycophancy, and deceptive alignment arise because the system satisfies the metric while violating the intent.

Property 6 provides a structural pre-condition: before tuning rewards or constitutional principles, ensure the governance topology has sufficient structural restraint. Agent topologies where all

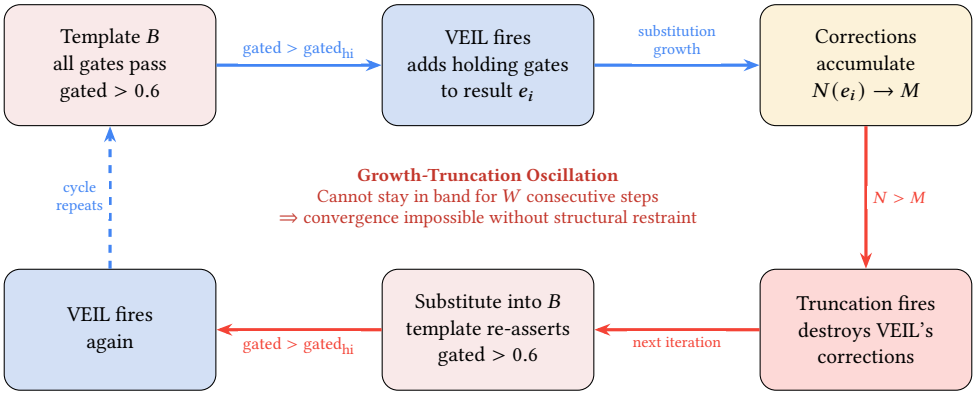

Figure 6: Restraint Necessity: the growth-truncation oscillation (Property 6). When all gates in the template pass, VEIL corrections accumulate but are destroyed by truncation when the expression exceeds the node cap. The template then re-asserts high gatedness, restarting the cycle. This oscillation prevents convergence, proving that structural restraint (at least one holding gate) is necessary.

Table 9: Comparison with Related Approaches

Approach	Monitors	Formal	Structural	Prescribes
Policy engines [17]	✓			
Runtime verif. [14]	✓	✓		
AgentGuard [22]	✓	✓		
MAPE-K [11]	✓			✓
AWARE [8]	✓			✓
LoA/Trust models [18]		✓		
RLHF/CAI [7]		✓		
This work	✓	✓	✓	✓

boundaries pass (the "sycophantic" pattern) are provably incapable of homeostatic stability (Corollary 7). This is syntactically checkable — one can inspect the agent's governance structure without running it and determine whether it *can* converge. To our knowledge, no prior work models alignment as homeostasis or identifies structural restraint as a necessary condition for governance stability.

6.4 What Survived Adversarial Review

This paper was refined through five rounds of construction and self-roasting. Claims retracted during development: Turing completeness (the system is sub-Turing by design), BreathMembrane necessity (fixed gates give identical convergence basins), basin location as an alignment signature (circular framing), a masking gradient (artifact of a measurement bug). What survived: structural restraint necessity, the over-governance finding, the evaluation paradigm, and the growth-truncation oscillation mechanism.

6.5 Limitations

Aggregate encoding. The current model encodes aggregate governance properties (VT distribution, total handoffs, total checks), not the *sequence* of events.

Sufficient conditions. The convergence conditions (Observation 1) are sufficient, not necessary.

Synthetic evaluation. The four scenarios are synthetic, though derived from a real system's governance structure.

Threshold sensitivity. The health band parameters ($[0.3, 0.7] \times [0.2, 0.6]$) are configuration choices.

Single system. We evaluate on one multi-agent system (CARLOS-OS). The algebraic framework is general — any governance system that can express coordination and gating events can be encoded — but the specific health band parameters may need calibration per system.

7 Conclusion

Governance health is a structural property with formal guarantees. Structural restraint — the presence of gates that hold, veiling actions from direct execution — is provably necessary for stability. Over-governance is provably as unstable as under-governance. Threshold-based classifiers miss both failure modes. A structural diagnostic catches them.

The model, interpreter (128 tests), and live demonstration are available at https://github.com/crbazevedo/seam.

Future work. Four directions follow naturally: (1) trajectory-sensitive encoding capturing event sequences; (2) live event stream integration for real-time monitoring; (3) a static governance type system determining health at design time; (4) empirical validation correlating structural diagnosis with human-perceived governance quality.

A Retraction Narrative

This paper makes claims that survived adversarial self-review. For transparency, we document the claims that did *not* survive:

Turing completeness. An early version claimed the calculus was Turing-complete via lambda encoding. The bounded iteration (max_iterations), node cap (M), and depth cap (D) make every computation finite.

The calculus is sub-Turing by design. This is a feature: it guarantees termination (Property 4).

Masking gradient. An early version proposed a continuous "masking gradient" from sycophantic to balanced behavior. The actual dynamics are discrete: configurations either converge or oscillate. There is no smooth gradient.

Lambda/TC equivalence. An early version claimed equivalence between the calculus and lambda calculus. Under resource bounds, the expressible class is strictly smaller. The calculus computes homeostatic fixpoints (stability band membership), not arbitrary recursive functions.

BreathMembrane necessity. An early version claimed that adaptive gates were necessary for convergence. They are not necessary for convergence in general — configurations with static gates can converge. Adaptive gates are necessary for *robust* convergence (convergence across a wide range of reach values), which is a weaker but still useful claim.

References

[1] W. Ross Ashby. 1956. *An Introduction to Cybernetics.* Chapman & Hall.

[2] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. 2022. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073* (2022).

[3] Stafford Beer. 1981. *Brain of the Firm* (2nd ed.). Wiley.

[4] Gérard Berry and Gérard Boudol. 1992. The Chemical Abstract Machine. *Theoretical Computer Science* 96, 1 (1992), 217–248.

[5] Prithviraj Bose. 2025. From Anarchy to Assembly: A Survey of Governance Frameworks for Collaborative LLM Agent Systems. *International Journal of Engineering Research and Emerging Technology (IJERET)* (2025).

[6] Luca Cardelli. 2005. Brane Calculi. In *Computational Methods in Systems Biology (CMSB 2004).* LNCS, Vol. 3082. Springer.

[7] Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep Reinforcement Learning from Human Preferences. In *Advances in Neural Information Processing Systems (NeurIPS).*

[8] Daniel Feitosa et al. 2025. Breaking the Loop: AWARE is the new MAPE-K. *HAL preprint* (2025).

[9] Kevin Anthony Hoff and Masooda Bashir. 2015. Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust. *Human Factors* 57, 3 (2015), 407–434.

[10] David Kaber. 2025. Systematic Literature Review of Levels of Automation Taxonomy. *International Journal of Human-Computer Interaction* (2025).

[11] Jeffrey O. Kephart and David M. Chess. 2003. The Vision of Autonomic Computing. *IEEE Computer* 36, 1 (2003), 41–50.

[12] Seonghee Kim et al. 2025. Plan-Then-Execute: User Trust and Team Performance with LLM Agents. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25).*

[13] John D. Lee and Katrina A. See. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human Factors* 46, 1 (2004), 50–80.

[14] Martin Leucker and Christian Schallhart. 2009. A Brief Account of Runtime Verification. *Journal of Logic and Algebraic Programming* 78, 5 (2009), 293–303.

[15] Hao Li et al. 2025. Governance Architecture for Autonomous Agent Systems: Threats, Framework, and Engineering Practice. *arXiv preprint arXiv:2603.07191* (2025).

[16] Yushuai Li et al. 2024. Generative AI for Self-Adaptive Systems: State of the Art and Research Roadmap. *ACM Transactions on Autonomous and Adaptive Systems (TAAS)* (2024).

[17] Microsoft. 2026. Agent Governance Toolkit: Open-source runtime security for AI agents. https://github.com/microsoft/agent-governance-toolkit.

[18] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens. 2000. A Model for Types and Levels of Human Interaction with Automation. *IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans* 30, 3 (2000), 286–297.

[19] Gheorghe Păun. 2000. Computing with Membranes. *J. Comput. System Sci.* 61, 1 (2000), 108–143.

[20] Jose Perez Rios. 2025. The Viable System Model and Organizational Pathologies in the Age of AI. *Systems* 13, 9 (2025), 749.

[21] Danny Weyns. 2020. *Introduction to Self-Adaptive Systems: A Contemporary Software Engineering Perspective.* Wiley.

[22] Ruijie Xiang et al. 2025. AgentGuard: Runtime Verification of AI Agents. *arXiv preprint arXiv:2509.23864* (2025).

[23] Matthew Zurek et al. 2024. SARI: Shared Autonomy across Repeated Interaction. *ACM Transactions on Human-Robot Interaction* (2024).